An Open-Source Tool for Lifetime Assessment of DC-Link Capacitors in Grid-Connected Converters

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***Abstract*—This paper presents the DC-link capacitor module developed within the PEARL (Power Electronics Age and Reliability) framework, an open-source Python based framework for mission profile based lifetime estimation of DC-link capacitors in grid-connected power electronic converters. The framework establishes a unified computational pipeline that links time-varying mission profiles, operating conditions, design configurations and component specifications to capacitor lifetime by integrating electrical stress modelling, loss calculation, thermal modelling and lifetime modelling. The implementation is modular and computationally efficient, enabling long duration simulations with direct datasheet-driven parameterization of DC-link capacitors. Case studies show that decreasing the power factor from unity to zero increases the RMS ripple current, total losses and core temperature, resulting in a reduction in the estimated capacitor lifetime from approximately 16 years to about 13.5 years. Furthermore, the ambient temperature analysis indicates that when the converter is placed in an ambient temperature of approximately 56 °C, the corresponding core temperature of the capacitor reaches about 99 °C, exceeding the allowable thermal limit and leading to capacitor failure due to thermal overstress. The results highlight the importance of reliability aware operation and design of DC-link capacitors in converter-dominated energy systems and position the presented framework as a tool for early-stage reliability assessment.**

***Index Terms*—Power electronics reliability, DC-link capacitor, electro-thermal modelling, mission profile, lifetime estimation**

I. INTRODUCTION

With the rapid expansion of converter-dominated electrical networks, the dependability of power electronic converters has become a decisive factor in maintaining stable grid operation. These converters act as the interface between distributed generation, storage units and the wider power system, making their reliability critical, while their failures contribute significantly to operational and financial losses in PV installations and remain among the leading causes of unplanned outages in wind turbines [1], [2]. Furthermore, inverter-based resources are being assigned responsibilities that extend far beyond simple power injection, including the provision of voltage support, reactive power control and ride through capability during network disturbances [3]. This expanding role increases stress on converter hardware. Therefore, with the continued growth and market participation of inverter-based resources [4], assessing the long-term reliability of key converter components is essential for planning, operation and investment decisions [1].

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Among converter components, DC-link capacitors are widely recognized as a major reliability concern because they contribute significantly to converter failure rate [1], [5], [6]. Their ageing is primarily governed by electrical and thermal stresses, with key stressors including ripple current, operating temperature, voltage stress, moisture and material degradation mechanisms [5], [6]. Since DC-link capacitors are essential for maintaining DC-bus voltage stability, absorbing power imbalance, filtering switching induced ripple and buffering transient energy fluctuations, their degradation can directly reduce converter performance, availability and service life [6].

A. Identified Research Gap

Although lifetime assessment of DC-link capacitors in grid connected power electronic converters is widely studied, a unified open-source framework for mission profile based lifetime assessment remains largely absent. Existing literature primarily presents isolated methodologies [5], [7], while available manufacturer-provided lifetime estimation tools (Table I) are typically closed-source, vendor-specific and limited to simplified operating scenarios. While some tools include electro-thermal modelling and design exploration features, they are generally restricted to discrete operating points rather than time-varying mission profiles representative of realistic operating conditions.

B. Paper Contribution

This paper presents the DC-link capacitor module of the open-source PEARL framework for mission profile based lifetime assessment of DC-link capacitors in grid-connected power electronic converters. The framework links time-varying mission profiles, operating conditions, converter design parameters and component specifications to component lifetime by integrating electrical stress modelling, loss calculation, thermal modelling and lifetime estimation within a unified, modular and computationally efficient workflow with direct datasheet-driven parameterization. In contrast to existing manufacturer-provided lifetime estimation tools(Table I), which are typically closed-source, vendor-specific and limited to simplified operating conditions, the proposed framework supports time-series mission profile analysis under realistic operating conditions, is technology agnostic, enables design exploration and provides a transparent and modifiable environment for lifetime assessment and reliability aware design of DC-link capacitors in grid-connected converters.

TABLE I: Comparison of manufacturer provided DC-link capacitor lifetime estimation tools and the proposed DC-link capacitor lifetime estimation framework. In this table, ✓ = fully supported, △ = limited capability, × = not supported.

Company	**Tool**	**Mission profile input**	**Mission profile lifetime**	**Unified electro-thermal-lifetime**	**Loss modelling**	**Thermal modelling**	**Technology agnostic**	**Design exploration**	**Modifiable/ Open**
TDK Electronics	CLARA	△	×	△	✓	✓	×	✓	×
Nichicon	Estimated Life Calculator	×	×	×	×	×	×	×	×
Cornell Dubilier	DC-link Film Life Calculator	×	×	×	×	△	×	△	×
Vishay	Lifetime Calculator	×	×	△	△	△	×	△	×
KEMET	WebLifeCalc	△	△	△	✓	✓	×	△	×
Kendeil	Life Calculation	△	×	△	✓	✓	×	△	×
Würth Elektronik	REDEXPERT	△	×	△	✓	✓	△	✓	△
This work	**PEARL**	✓	✓	✓	✓	✓	✓	✓	✓

C. Paper Organization

Section II presents the proposed methodology, Section III describes the case studies and Section IV discusses the corresponding results and key findings. Finally, Section V concludes the paper, while Section VI outlines directions for future work.

II. Methodology

This section describes the modelling workflow used for lifetime estimation of the DC-link capacitors in grid-connected converters, including electrical stress modelling, loss calculation, thermal modelling, and lifetime assessment.

The analysis begins by determining the electrical stresses imposed on the DC-link capacitor under a given converter mission profile, defined by the active power, reactive power and AC-side voltage at the converter terminals, together with the DC-link voltage and other required system parameters. From these quantities, the apparent power, inverter current, power factor and phase angle are obtained using standard power flow relations. These electrical quantities are then used to calculate the RMS ripple current flowing through each DC-link capacitor using an analytical expression based on the formulation presented in [8].

The voltage and current obtained in the previous step are used to compute capacitor losses arising from ripple current conduction P_{ripple} and dielectric leakage P_{leak}, with the corresponding expressions given in (1), where I_{rms} is the RMS ripple current flowing through each DC-link capacitor, ESR_{eff} denotes the effective series resistance, R_{leak} the leakage resistance and V_{cap} the applied capacitor voltage.

$$P_{\text{ripple}} = I_{\text{rms}}^2 \cdot \text{ESR}_{\text{eff}}, \qquad P_{\text{leak}} = \frac{V_{\text{cap}}^2}{R_{\text{leak}}} \tag{1}$$

The total losses obtained from the loss modelling stage represent the internal heat generation within the capacitor and serve as the input to the thermal model to estimate the resulting temperature rise. A lumped-parameter single-node steady-state thermal model, as defined in (2), is used to compute the capacitor core temperature, where T_{core} is the core temperature, T_{amb} is the ambient temperature, k is a calibration factor, R_{th} is the thermal resistance and P_{loss} denotes the total internal power dissipation.

$$T_{\text{core}} = T_{\text{amb}} + k\, R_{\text{th}}\, P_{\text{loss}} \tag{2}$$

A calibration factor k is introduced in (2) to improve the accuracy of the thermal model. It is typically equal to unity but may take higher values e.g., $k = 1.5$ for aluminum electrolytic capacitors from Cornell Dubilier to account for internal construction effects [9]. When available, k can be obtained from the manufacturer datasheet otherwise, it can be determined by matching the modelled temperature rise to the rated temperature rise under reference conditions. Furthermore, it should be noted that (2) represents a simplified lumped-parameter steady-state thermal model. Although more detailed thermal models, such as Foster or Cauer networks, can provide higher accuracy, they require thermal time constants and internal parameters that are not consistently available in manufacturer datasheets for DC-link capacitors. In addition, capacitor ageing is primarily governed by the average core temperature due to their relatively slow thermal dynamics, hence, a dynamic thermal model is generally not required. Therefore, a simplified steady-state model is adopted to enable datasheet-driven modelling and maintain computational efficiency for long mission profile simulations.

With the capacitor core temperature and voltage determined for each set point of the mission profile, the corresponding capacitor lifetime can be evaluated. The framework integrates various lifetime models from manufacturers such as Nichicon, Rubycon, Panasonic, Faratronic and Cornell Dubilier, covering the two predominant DC-link capacitor technologies, namely aluminum electrolytic and film capacitors. In addition to these specific models, the methodology incorporates a generalized Arrhenius type voltage temperature acceleration model, alongside a novel graph-based lifetime model developed by the author that enables the direct application of lifetime curves extracted from manufacturer datasheets. Because the presented framework is designed with a highly modular architecture, any new lifetime model can be seamlessly integrated without necessitating modifications to the overall system. Finally, the lifetime corresponding to each mission profile set point is calculated and then Miner's rule is applied to accumulate the damage and estimate the expected lifetime of the DC-link capacitor.

Fig. 1 provides a visual overview of the framework pipeline. Detailed descriptions of the implemented models are provided in the document *Model_documentation_PEARL_DC_Link_Capacitor.pdf* located in the *DC Link Capacitor* folder of the PEARL GitHub repository [10], covering the electrical stress, loss modelling, thermal and lifetime assessment models. In practice, the framework only requires a small set of readily available inputs, namely the capacitor manufacturer datasheet parameters, the inverter configuration and the time-varying active and reactive power profiles along with ambient temperature.

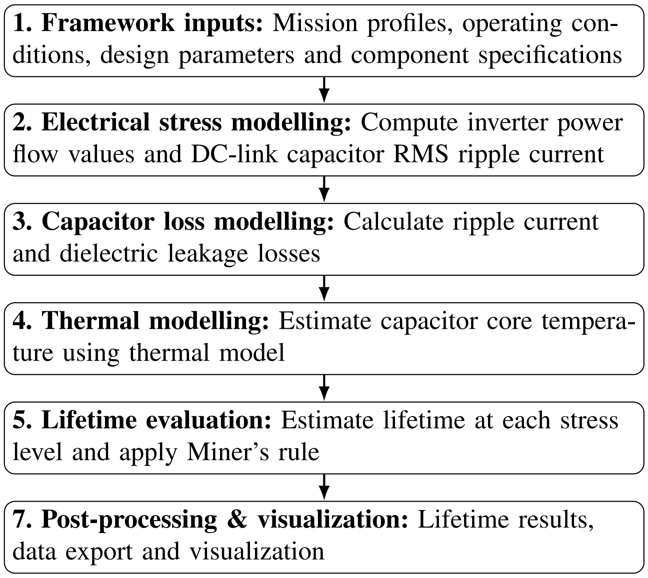

Fig. 1: Methodological workflow.

III. Case Study

A. Case Study 1: Reactive power impact

Motivated by the increasing requirement for power converters to provide reactive power support [3], this case study quantifies its impact on the lifetime of DC-link capacitors installed in grid-connected converters. The setup, shown in Fig. 2, considers a three-phase inverter connected to the grid supplying active and reactive power, operating under a fluctuating apparent power profile (Fig. 3). The apparent power profile is synthetically generated as a bounded stochastic signal around 1 MW with piecewise-constant 15-minute intervals to emulate realistic inverter loading conditions, since detailed long-term mission profiles of power converters are rarely publicly available. The mission profile spans one year (with one representative day shown for clarity), during which the inverter operates at constant power factor values between 0 and 1 under both inductive and capacitive conditions, resulting in 20 operating scenarios. For each scenario, the corresponding operating conditions are cyclically repeated over the mission profile duration to evaluate the lifetime impact. The simplified scenario is used to isolate the effect of reactive power and does not limit the framework, which is capable of handling realistic long duration mission profiles with time-varying operating conditions.

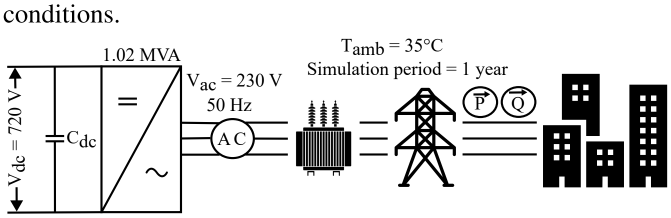

Fig. 2: Case study framework

In Fig. 2, the inverter is modeled as a three-phase converter rated at 1.02 MVA with time-series AC-side phase RMS voltage and DC-link voltage profiles, which are assumed constant at 230 V and 720 V, respectively, operating at a modulation index of 1 using space vector modulation (10 kHz

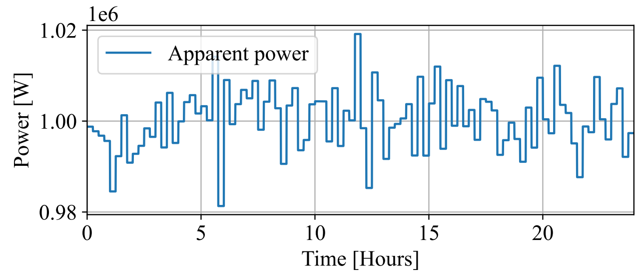

Fig. 3: Apparent power

switching, 50 Hz grid). The DC-link employs 11 parallel film capacitors (TDK B32778J8277K000, 800 V, 70.5 A, 270 μF) [11], selected to meet voltage and ripple current requirements. Key electrical, thermal and lifetime parameters are extracted from the datasheet, with all values provided in *Input_parameters.py* in the PEARL repository [10]. Lastly, the ambient temperature, which is treated as a time-series input in the framework, is assumed constant at 35°C.

B. Case Study 2: Ambient temperature impact

This case study investigates the influence of ambient temperature on the lifetime of the DC-link capacitor in grid-connected power electronic converters. The time-varying mission profile, operating conditions, design configuration and component specifications remain the same as in Case Study 1 (Section III-A), with the only differences being that the DC-link voltage is set to 560 V to improve visualization of the results and the power factor is fixed at $pf = 0$, thereby keeping the electrical stress conditions constant, namely the DC-link voltage, capacitor RMS ripple current, and the resulting power losses. Since the calibration factor k and thermal resistance R_{th} are constant and the electrical stress conditions are fixed, the power loss P_{loss} in (2) also remains constant. Therefore, the ambient temperature T_{amb} becomes the dominant variable governing the capacitor core temperature T_{core} and thus the capacitor lifetime, meaning that the variation in core temperature and lifetime can be directly attributed to changes in ambient temperature.

IV. Results and Discussion

Fig. 4 illustrates the temporal evolution of the DC-link capacitor RMS ripple current, total power losses and core temperature for the conditions defined in Case Study 1 (Section III-A). The results are shown at unity power factor, with a single day presented for clarity. This figure is included for illustrative purposes only to demonstrate the electro-thermal outputs generated by the proposed DC-link capacitor module of PEARL and does not represent the results of Case Study 1.

Fig. 4(a) shows the RMS ripple current per DC-link capacitor varying with the apparent power profile in Fig. 3, which leads to corresponding variations in internal power dissipation as shown in Fig. 4(b), governed by (1). Since the DC-link voltage is constant at 720 V (Case Study 1 Section III-A), the leakage losses P_{leak} remain constant, while the ripple losses P_{ripple} depend on the RMS current therefore, the total losses are primarily driven by the current. These losses represent the internal heat generation within the capacitor and serve as the input to the thermal model defined in (2). Since the ambient temperature T_{amb}, calibration factor k and thermal resistance R_{th} are constant, the total losses dominate the core temperature. Consequently, the core temperature follows the same trend as the total power losses, as shown in Fig. 4(c), reflecting the variations in the apparent power profile.

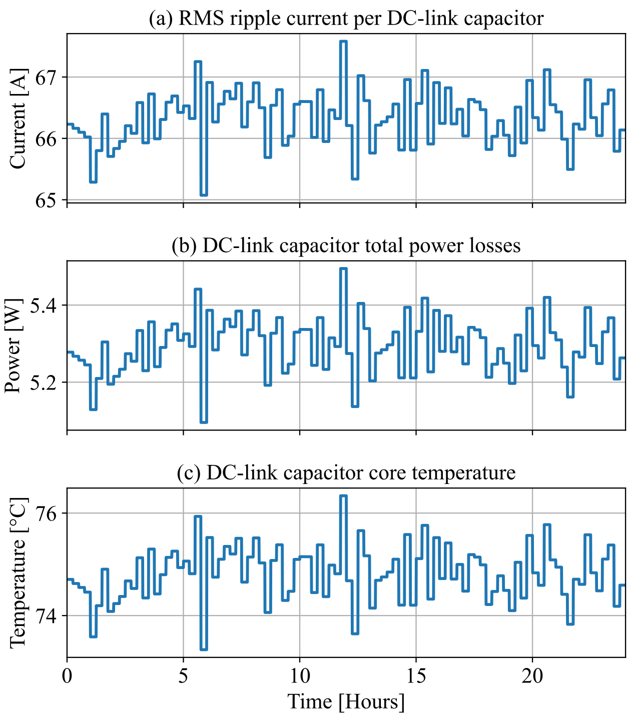

Fig. 4: (a) RMS ripple current, (b) total power losses, and (c) core temperature per DC-link capacitor at power factor 1.

A. Case Study 1 Results: Reactive power impact

Fig. 5 summarizes the impact of reactive power operation on the DC-link capacitor for the scenarios defined in Case Study 1 (Section III-A). The average RMS ripple current, total power losses, core temperature and the corresponding estimated lifetime are evaluated and compared for each scenario.

Fig. 5(a) illustrates the average RMS ripple current per DC-link capacitor, which increases as the power factor decreases from unity. At $pf = 1$, a baseline ripple current of about $66A$ exists due to converter switching, where the pulsating current drawn from the DC source is buffered by the DC-link capacitor. As the power factor decreases toward $pf = 0$, the phase shift between voltage and current increases, resulting in increased reactive power exchange. Consequently, energy is periodically stored and returned by the AC side rather than being continuously delivered to the load. This increases the power mismatch between the relatively constant DC input power and the AC-side power, which must be balanced by the DC-link capacitor. As a result, the RMS ripple current increases to approximately $69A$ [8].

As the RMS ripple current increases with decreasing power factor, the DC-link capacitor losses increase from approximately $5.25\,\mathrm{W}$ at $pf = 1$ to about $5.75\,\mathrm{W}$ at $pf = 0$, resulting in higher total power dissipation as shown in Fig. 5(b). Since capacitor losses dominate the thermal behaviour, the core temperature follows the same trend, increasing from approximately $75°\mathrm{C}$ to slightly above $78°\mathrm{C}$ as the power factor decreases from unity to zero, as illustrated in Fig. 5(c). The relationships between ripple current and total losses, as well as between losses and the thermal response of the capacitor core, have already been discussed in the explanation of Fig. 4 at the beginning of Section IV.

Although several factors influence DC-link capacitor ageing, the dominant ageing variables are core temperature and operating voltage. This is particularly evident for the TDK B32778J8277K000 film capacitor, whose manufacturer datasheet shows lifetime primarily as a function of temperature and voltage [11]. In Case Study 1 (Section III-A), the DC-link voltage remains constant at 720 V; therefore, temperature becomes the dominant ageing factor governing capacitor lifetime. Consequently, the temperature trend observed in Fig. 5(c) directly explains the lifetime reduction shown in Fig. 5(d), where the lifetime decreases from slightly above $16\,\mathrm{years}$ at $pf = 1$ to approximately $13.5\,\mathrm{years}$ at $pf = 0$. Furthermore, Fig. 5 shows that the direction of reactive power (capacitive or inductive) has no noticeable influence on the electro-thermal behaviour of the DC-link capacitor and consequently does not affect the estimated lifetime. These results highlight the usefulness of the DC-link capacitor lifetime assessment module developed within the PEARL framework for evaluating the impact of operating conditions, such as reactive power provision on capacitor lifetime during the early design stage of power electronic converters.

B. Case Study 2 Results: Ambient temperature impact

Fig. 6 summarizes the impact of ambient temperature on the capacitor thermal behaviour and lifetime. Fig. 6(a) illustrates the effect of ambient temperature on the estimated lifetime of the DC-link capacitor, while Fig. 6(b) shows the corresponding core temperature, demonstrating how changes in ambient temperature propagate to the capacitor core temperature and consequently affect capacitor lifetime. The ambient temperature is initially set to $0°\mathrm{C}$, resulting in a core temperature of approximately $43°\mathrm{C}$ and an estimated lifetime of $37.18\,\mathrm{years}$, corresponding to the maximum lifetime specified for the TDK Electronics film capacitor B32778J8277K000 at 0.7 of the rated voltage (560 V) [11]. As the ambient temperature increases, the lifetime initially remains at its maximum value because the core temperature stays within the rated operating range. However, when the ambient temperature reaches approximately $25°\mathrm{C}$, the core temperature rises to about $70°\mathrm{C}$ and the lifetime begins to decrease due to thermally accelerated ageing mechanisms such as dielectric degradation and insulation ageing [6]. At higher ambient temperatures, the core temperature continues to increase, and once it approaches approximately $99°\mathrm{C}$ at an ambient temperature of about $56°\mathrm{C}$,

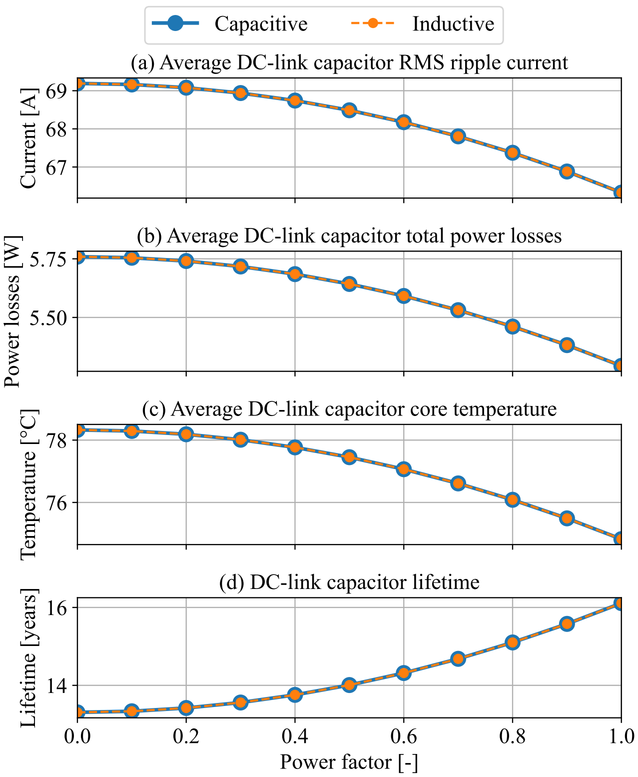

Fig. 5: Impact of reactive power on DC-link capacitor: (a) average RMS ripple current, (b) average total power losses, (c) average core temperature and (d) estimated lifetime.

the allowable thermal limit is exceeded, resulting in a rapid reduction of the expected lifetime. Overall, Fig. 6 demonstrates the strong influence of ambient temperature on the lifetime of the DC-link capacitor and highlights the importance of the presented DC-link capacitor module developed within the PEARL framework, which can support initial design of DC-link capacitor bank within grid-connected power electronic converters by enabling the evaluation of thermal stress and lifetime under different ambient temperature conditions.

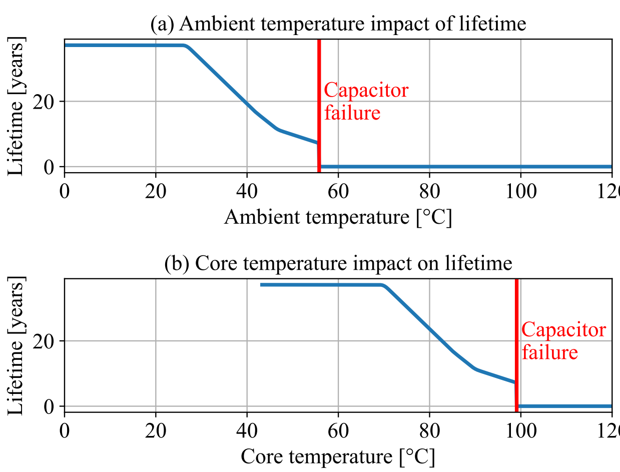

Fig. 6: Impact of (a) ambient temperature and (b) core temperature on capacitor lifetime

V. Conclusion

This paper presented the DC-link capacitor module developed within the open-source PEARL framework for mission profile based lifetime estimation of DC-link capacitors in grid-connected power electronic converters. The case studies demonstrate that converter operating conditions, particularly reactive power provision and ambient temperature, significantly influence capacitor thermal stress and lifetime. Reactive power operation increases RMS ripple current, losses and core temperature, reducing the estimated capacitor lifetime from approximately 16 years to 13.5 years, while higher ambient temperatures accelerate ageing, reducing the expected lifetime from approximately 37 years to about 5 years, with capacitor failure occurring as the core temperature approaches the allowable thermal limit of approximately 99°C. These results demonstrate that the presented framework provides a practical tool for reliability aware design by enabling an early stage evaluation of operating conditions and their impact on the lifetime of DC-link capacitor in grid-connected converters. Furthermore, the presented methodology is scalable and not limited to grid-connected converters, and can be extended to a wide range of inverter systems, including residential converters, microgrids and islanded power systems, enabling broader reliability assessment across converter-based energy systems.

VI. Future Outlook

Future work will extend the framework to additional power electronic components enabling PEARL to be expanded toward a system level reliability assessment of grid-connected power converters.

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